Computer Networks: Homework-2

Rudransh Verma
230881
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Solutions

Fig. 1 Based Questions

1. **Within the IP packet header, what is the value in the upper layer protocol field?**

   The value in the Protocol field is: **1 (ICMP)**.This is determined from the IP header where "Protocol: ICMP (0x01)" is shown.

2. **How many bytes are in the IP header? How many bytes are in the payload of the IP datagram? Explain how you determined the number of payload bytes.**

   Header Length = 20 bytes

   This is determined from:

   $$\text{Header Length field} = 5 \quad \text{(in 32-bit words)}$$

   $$5 \times 4 = 20 \text{ bytes}$$

   Total Length = 56 bytes

   Payload Size:

   $$56 - 20 = \textbf{36 bytes}$$

3. **Has this IP datagram been fragmented? Explain how you determined whether or not the datagram has been fragmented.**

   No, the datagram has NOT been fragmented because:

   - Flags = 0x00
   - More Fragments (MF) bit = 0
   - Fragment Offset = 0

   Since MF = 0 and Fragment Offset = 0, this packet is not fragmented.

4. **What is the value in the Identification field and the TTL field?**

   Identification: **0x0d02** (3326 in decimal)

   Time To Live (TTL): **1**

Fig. 2 Based Questions

5. **Can you say whether the message corresponding to the above packet has been fragmented?**

   Yes, the message has been fragmented because the Info column explicitly shows "Fragmented IP protocol", and the Flags field indicates fragmentation.

6. **What information in the IP header indicates that the datagram been fragmented?**

   The following fields indicate fragmentation:

   - Flags field: More Fragments (MF) bit = 1
   - Fragment Offset field

   In Fig. 2, Flags = $0x02 \Rightarrow$ MF bit set

7. **What information in the IP header indicates whether this is the first fragment versus a latter fragment?**

   The **Fragment Offset field** determines this:

   - First fragment: Fragment Offset = 0
   - Later fragment: Fragment Offset > 0

   In Fig. 2, Fragment Offset = 0. Hence, this is the first fragment.

Fig. 3 Based Questions

8. **What information in the IP header indicates that this is not the first datagram fragment?**

   Because **Fragment Offset** = 1480. Since Fragment Offset > 0, this is not the first fragment.

9. **Are the more fragments? How can you tell?**

   No, because Flags = 0x00. The More Fragments (MF) bit is not set, which means this is the last fragment.

10. **If Fig. 2 and Fig. 3 are the 1st and 2nd fragments of a message, then what fields change in the IP header between the first and second fragment?**

    The following IP header fields change:

    - Total Length: 1500 → 520
    - Fragment Offset: Set → Not set
    - Flags (MF bit changes): 0 → 1480
    - Header Checksum (recalculated): 0xda69 → 0xfd84